TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Subhajeet Rath	Email	subhajeet.rath@tno.nl
Institute Name	Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO), Powertrains Department			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	AI-enabled battery digital twins and charging orchestration for large-scale EV fleet operations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical / Electronics		Information / Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Software	SW Solutions
Concept of Proposed Technology	Large-scale EV fleet operators face simultaneous constraints on vehicle availability, charger availability, electricity cost, and grid capacity. This proposal targets an integrated software stack that combines: 1) vehicle and battery digital twins for charge-time prediction and battery health estimation, 2) real-time, constraint-aware charging orchestration to minimize total cost of ownership by balancing energy cost and battery degradation, and 3) fleet-level KPIs (uptime, energy use, carbon accounting inputs) exposed through APIs for integration into an existing EV fleet operations platform of the Korean SME partner. TNO will contribute model-based and data-driven methods for battery digital twins and fleet			

Rationale for Joint R&D Project	charging optimization that have been published and demonstrated in European projects and peer-reviewed venues (2)-(6).
	The Korean SME partner should have an operational EV fleet management platform and access to fleet telemetry, charging-session data, and user requirements from commercial deployments (partner-provided, non-public). However, next-generation capabilities such as charge-time digital twins, battery health estimation, and degradation-aware charging optimization require specialized electrochemical modeling, system identification, and validation methodologies. TNO has published methods on battery digital twins for charge-profile and charge-time prediction (4) and real-time fleet charging optimization that explicitly accounts for battery degradation and grid constraints (3) and (5). Additionally, TNO develops system-level interfaces and requirements for operational fleet support tools including digital twins and fleet planning, with attention to interoperability with energy management systems and charge station management systems (5). International joint R&D enables a faster, lower-risk path to integrate validated algorithms into the SME platform, pilot them in relevant operational environments, and generate jointly-owned IP suitable for commercialization.
Key Words	1. Battery digital twin 2. Degradation-aware charging orchestration 3. Fleet energy management 4. Battery health monitoring (SoH, anomaly detection) 5. API-based platform integration

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	Develop and validate an end-to-end algorithmic stack for EV fleet energy management that is deployable in an operational SaaS context: 1) Battery digital twin for charge-time prediction and state estimation (SoC, SoH) using fleet data. 2) Fleet charging optimization that produces per-vehicle power profiles under grid, charger, and schedule constraints, including explicit battery degradation costs. 3) Anomaly detection and alerting at the charging-session level (data-quality checks, outlier detection, and early fault indicators) to support operational decision-making. 4) Integration of the above modules through secure APIs into the SME partner platform, enabling pilot trials with fleet operators.			
R&D Target Specifications	Target specifications to be validated on pilot fleet data (values are project targets): • Charge-time prediction: <2% error on charge time for representative charging sessions (baseline improvement consistent with prior published results) (4). • Fleet charging optimization: Demonstrate measurable reduction in total charging cost (energy + degradation proxy) relative to baseline heuristics; reference performance includes 13% simulated savings reported in published work (5). • Runtime: Scheduling computation <3 minutes for 200 vehicles (order of magnitude aligned with published real-time results) (5). • Interoperability: API interfaces compatible with common fleet and charging system concepts (digital twin, fleet planning, EMS, CSMS) as specified in TNO-led requirements work (6).			
R&D Plan	Proposed work packages (example 18-month plan; exact schedule to align with TIPA track): **WP1. Requirements and data onboarding (Months 1-3)** - Define use cases, KPIs, and data schemas; map SME telemetry and charging logs to a common data model. - Establish privacy and security controls and evaluation datasets. **WP2. Battery digital twin development (Months 2-9)** - Implement charge-profile and charge-time prediction models; calibrate using fleet data. - Validate prediction performance and uncertainty bounds. **WP3. Degradation-aware charging optimization (Months 4-12)** - Implement fleet-level scheduling optimizer that accounts for grid and depot constraints and battery degradation costs. - Benchmark against baseline scheduling policies. **WP4. Operational anomaly monitoring (Months 6-14)** - Develop anomaly detection on charging sessions and vehicle energy signals; integrate alert			

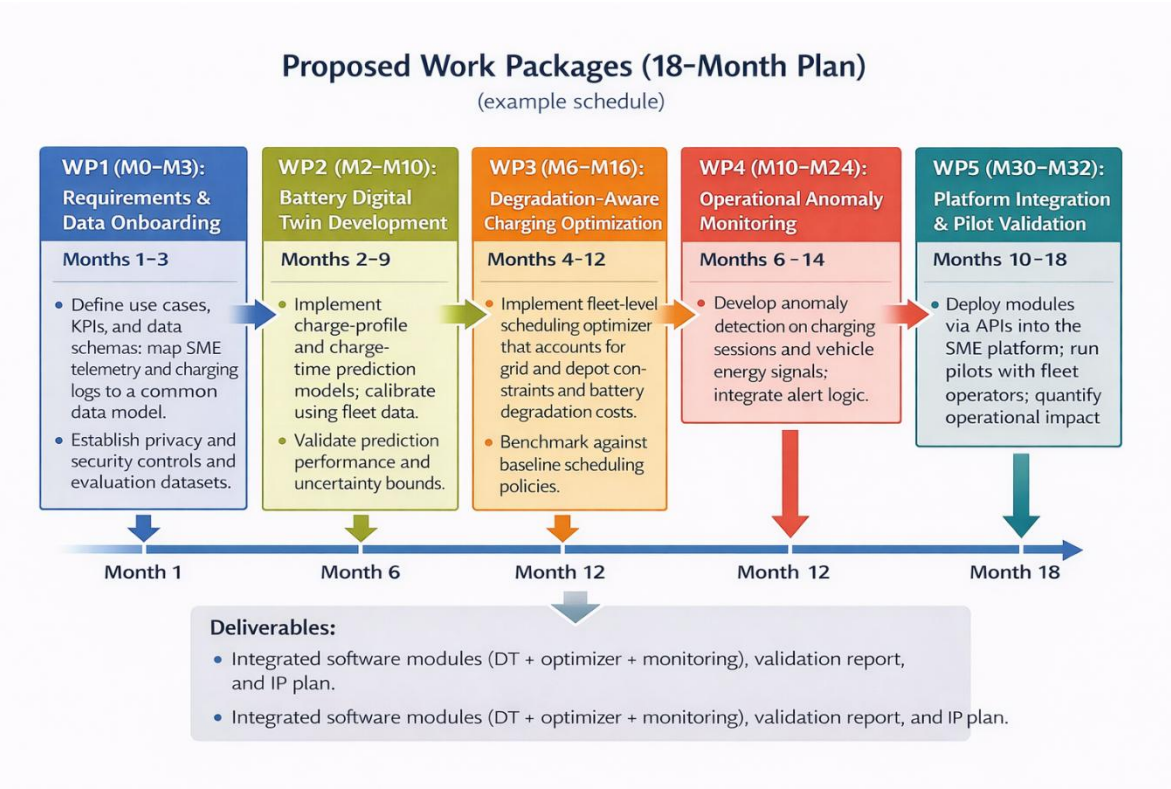
logic.

WP5. Platform integration and pilot validation (Months 10-18)

- Deploy modules via APIs into the SME platform; run pilots with fleet operators; quantify operational impact.

Deliverables

- Integrated software modules (DT + optimizer + monitoring), validation report, and IP plan.



4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (research institute) expected roles: • Technical lead for battery digital twin modeling and validation methods (4),(6). • Develop and prototype degradation-aware charging optimization and real-time scheduling algorithms (6). • Define reference architectures and API interfaces for digital twin and fleet planning integration (6). • Provide project management support, scientific reporting, and contribution to IP strategy. • Provide domain expertise in vehicle energy management including Modular Energy Management (MEM) methods for efficient onboard control implementation (2).
Expected R&R of Korean SME Partner	Korean SME partner expected roles: • Provide operational requirements, deployment constraints, and access to representative fleet and charging-session datasets (partner-provided, non-public). • Provide integration endpoints and engineering resources for incorporating TNO modules into the SME platform (data ingestion, API gateway, UI/UX). • Arrange pilot customers and field validation sites; support operational monitoring and feedback loops. • Lead commercialization planning in Korea and target export markets, including productization and customer support.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: • A validated battery digital twin module for charge-time prediction and battery state estimation suitable for integration into fleet operations software (4),(6). • A real-time fleet charging orchestration module that reduces operational charging costs and manages battery degradation under grid constraints (3) and (5). • Operational anomaly monitoring for charging sessions and vehicle energy signals to support reliability and safety management. • Pilot demonstrations with quantified KPIs and a commercialization plan for deployment within the SME product. • Jointly generated intellectual property and publications, aligned with TIPA requirements. Broader impact: • Improved EV fleet uptime and reduced total cost of ownership through optimized energy management. • Support for scalable electrification of logistics fleets by enabling digital tools that integrate vehicle planning, digital twins, and charger control (6).

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization